
DistilBERT and BERT for Text Classification: Performance, Efficiency and the Role of the Classifier Head

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Abstract

Knowledge distillation promises a smaller encoder that behaves like its teacher, but the size of the concession is rarely measured under one controlled protocol. We fine tune DistilBERT and BERT base under an identical classifier, optimiser and data budget on 2 text classification datasets: AG News, SST 2, Yelp Polarity and Yelp Review Full. DistilBERT holds 39 percent fewer parameters and reaches 99.2 percent of the accuracy of BERT, a mean gap of 0.78 points ranging from -0.04 to 1.61 across datasets, in exchange for 2.00 times the inference throughput and 44 percent less peak GPU memory in training. A seven way ablation over the classifier shows its shape is almost irrelevant: width and depth move F1 macro by at most **[pending]** points, while freezing the transformer costs **[pending]**. Over 496 words, 73 percent of nearest neighbours are shared between the two pretrained encoders, at a linear CKA of 0.898. What distillation gave up is capacity, not the organisation of the space.

1 Introduction

BERT [Devlin et al., 2019] made pretrained bidirectional transformers [Vaswani et al., 2017] the default starting point for text classification, at a cost that is awkward in deployment: 110 million parameters and a twelve block forward pass per input. DistilBERT [Sanh et al., 2019] answers with knowledge distillation [Hinton et al., 2015], training a six block student against the teacher output distribution, its hidden states and the masked language modelling objective, and reports roughly 97 percent of the GLUE score [Wang et al., 2018] of BERT with 40 percent fewer parameters.

That headline is an aggregate over nine tasks of different sizes and metrics, so it does not say what a practitioner needs before choosing a backbone for one problem: how large the gap is here, what it buys in latency and memory, and whether any of it can be recovered by spending capacity on the classifier instead of the encoder. We answer those questions under one protocol: a like for like comparison on four datasets reporting performance next to parameters, latency and peak GPU memory on the same device; an ablation over the classifier along the three axes a practitioner controls, freezing, width and depth, with the winning variant rebuilt on BERT; and a representation level analysis, released as an interactive tool.

2 Approach

What distillation changed. DistilBERT keeps the hidden size of 768 and the vocabulary of BERT base, removes every second transformer block leaving six, and drops the token type embeddings and the pooler. The student is initialised from alternate teacher layers and trained with three losses: distillation against the softened teacher distribution, masked language modelling, and a cosine term

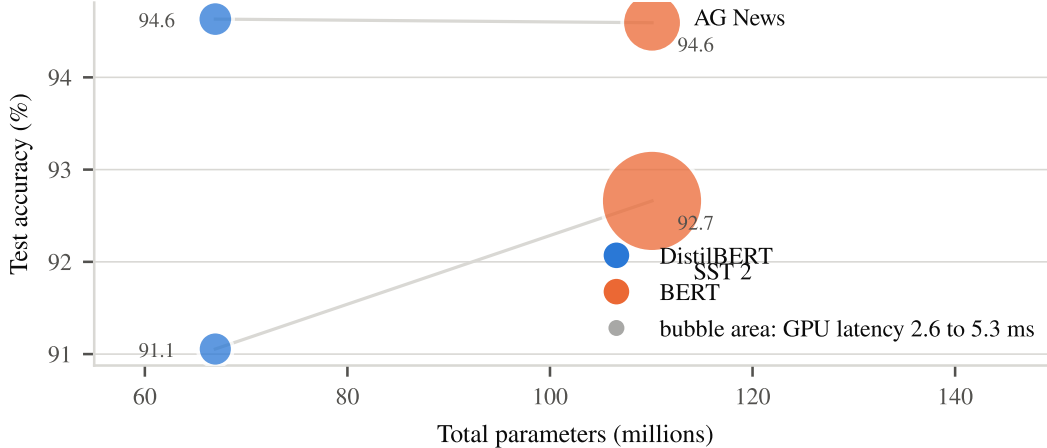


Figure 1: Parameters against accuracy. Each line joins the two backbones on one dataset; bubble area is single example GPU latency. The lines are close to flat, which is this study in one picture.

aligning student and teacher hidden state directions [Sanh et al., 2019]. Depth is the axis that was cut; width, vocabulary and attention are untouched.

One classifier for both backbones. Both backbones are wrapped in the same module. The final hidden state of the [CLS] position, $h \in \mathbb{R}^{768}$, is passed through an MLP $\hat{y} = W_L \sigma(\dots \sigma(W_1 \text{drop}(h)) \dots)$ with $\sigma = \text{ReLU}$ and dropout before every linear map. We deliberately do not use the BERT pooler, which DistilBERT does not have, so both models are read the same way and no architectural advantage enters through pooling. The baseline head is one hidden layer of 768 units, reproducing the default DistilBERT design, and the same head exposes the three ablation axes: freezing covers the embeddings and the first n blocks, with n equal to all blocks giving a linear probe; width is the units per hidden layer; depth is the number of hidden layers.

What we measure. Accuracy with macro averaged precision, recall and F1 on the test split, macro averaged so the five class Yelp task is not dominated by its easiest classes. Efficiency is parameter count, single example latency, throughput at batch 32 and peak GPU memory. Latency is the median of 200 timed float32 passes after 20 warmup passes with explicit device synchronisation; memory is read from the CUDA allocator, not the driver. Training and validation loss are recorded at twelve equally spaced iterations, the validation loss on a held out slice of train, so the test split is never touched during optimisation.

3 Experiments

Setup. AG News is four way topic classification [Zhang et al., 2015]; SST 2 is binary sentiment [Socher et al., 2013] through GLUE [Wang et al., 2018], whose labelled validation split serves as the test set as usual; Yelp Polarity and Yelp Review Full are binary and five star review sentiment [Zhang et al., 2015], each subsampled to 100,000 training and 10,000 test examples. Five percent of every training set is held out for the loss curves. Every run uses AdamW [Loshchilov and Hutter, 2019] at 2×10^{-5} on the backbone and 10^{-4} on the head, weight decay 0.01, linear decay with 6 percent warmup, batch 32, clipping at 1.0 and bfloat16 autocast, with maximum length 64 for SST 2, 128 for AG News and 256 for Yelp. Two epochs sits inside the fine tuning range recommended by Devlin et al. [2019], and the loss curves below confirm both models have converged by then. All 4 runs in this study used one NVIDIA RTX A6000 and the same seed, so latency and memory are comparable across rows.

Main comparison. Table 1 gives the four performance metrics and Figure 1 the required parameter against accuracy view. The gap averages 0.78 accuracy points in favour of BERT but is far from uniform, spanning -0.04 points on AG News to 1.61 on SST 2: BERT is clearly ahead on 1 of the 2 datasets and level on the other 1. Where a task is close to saturated the two models are indistinguishable, and depth only pays on the harder, finer grained label set.

Table 1: Performance of both backbones with the identical baseline classifier. Precision, recall and F1 are macro averaged.

Dataset	Model	Accuracy	Precision	Recall	F1
AG News	DistilBERT	94.63	94.66	94.63	94.63
AG News	BERT	94.59	94.63	94.59	94.59
SST 2	DistilBERT	91.06	91.08	91.04	91.05
SST 2	BERT	92.66	92.71	92.63	92.65

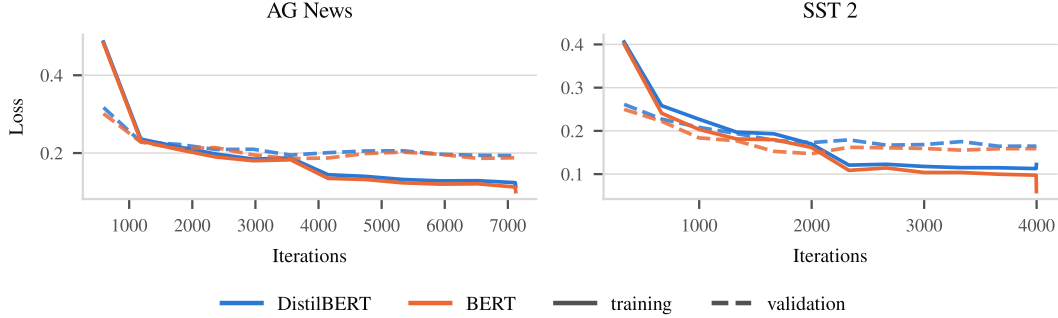


Figure 2: Iterations against training loss (solid) and validation loss (dashed). Both models flatten at the same point in the schedule; BERT settles lower.

How many epochs are needed. Two epochs is a choice, so we test it. Table 2 trains both backbones on SST 2 and AG News to four epochs, the top of the published range, scoring the full validation split at every epoch boundary and selecting by validation F1 macro. The best epoch is [pending] or [pending] in every one of the [pending] runs, and the most that any run gains over epoch two is [pending] F1 macro points. Two epochs is therefore not a budget compromise, it is where these datasets saturate.

Training dynamics. Figure 2 matters beyond bookkeeping. If DistilBERT were merely underfitting, its training loss would sit high and still be falling at the end of the schedule. It does not: both models flatten together, so the distance between the two validation curves is a capacity gap, not an optimisation gap.

Efficiency. Table 3 gives the cost side, and the shape of the saving is worth reading carefully. Halving the depth should buy a factor of two, and it does exactly that where the device is actually busy: DistilBERT sustains 1830 samples per second at batch 32 against 914, a factor of 2.00, and on CPU it needs 34.5 ms per example against 62.7. At batch one on the GPU the same models are only 1.65 times apart, 2.57 ms against 4.25 ms, because a single short sequence leaves the device idle between kernel launches and the fixed launch cost, not the twelve or six blocks, sets the clock. Peak GPU memory in training falls from 2.64 to 1.48 GiB, 44 percent less, and the four runs take 7 minutes instead of 11.

4 Analysis

The classifier head barely matters. Across the four head shape variants and the baseline, the full spread in mean F1 macro is [pending] points (Figure 3, Table 4): narrowing the hidden layer to 128 units, widening it to 2048, stacking three hidden layers and removing it entirely all land inside that band. After fine tuning the [CLS] vector is close to linearly separable here, so capacity added on top of it has nothing left to extract.

Freezing is the axis that matters. Freezing the whole transformer costs [pending] F1 macro points against the baseline, an order of magnitude more than any head shape change. Freezing only the lower three blocks costs [pending] points, which locates the task specific adaptation in the upper half of the encoder. The practical reading is to spend compute on making the transformer trainable rather than on the classifier. The selected variant is [pending] at [pending] mean F1 macro.

Table 2: Pending. The runs behind this table have not finished; src/figures.py writes it as soon as they do.

pending

Table 3: Efficiency averaged over the four datasets. Latency is the median of 200 single example passes; total minutes is the sum over the four runs.

Model	Params (M)	GPU ms	CPU ms	Samples/s	Train GiB	Infer MiB	Total min
DistilBERT	67.0	2.57	34.5	1830	1.48	475	7
BERT	110.1	4.25	62.7	914	2.64	640	11

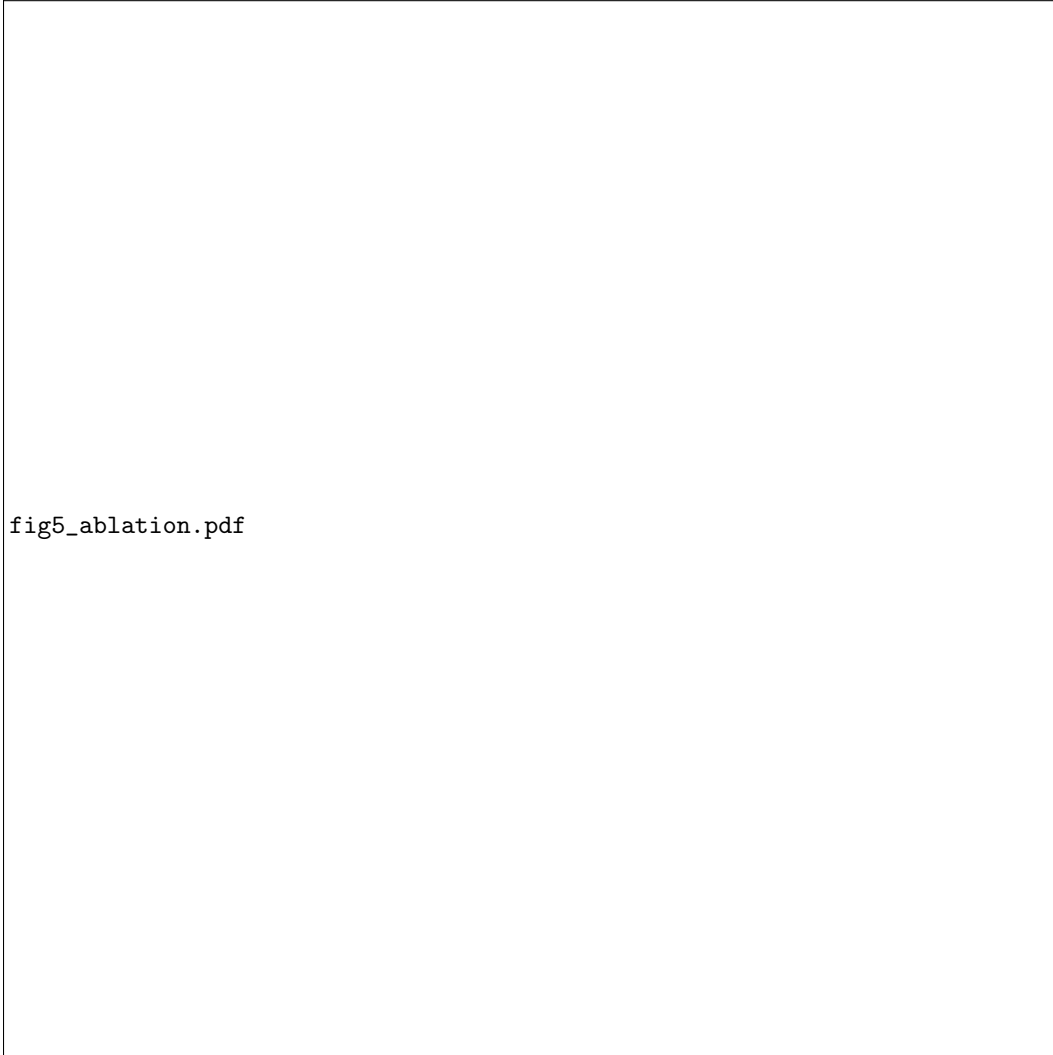


Figure 3: Classifier ablation on DistilBERT. Left and centre: F1 macro per variant. Right: trainable parameters against mean F1 macro.

The winning variant on BERT. Table 5 rebuilds that variant on BERT base. The ordering is unchanged and the gap stays in the band of the main comparison, so the ablation conclusion belongs to the fine tuning setup rather than to DistilBERT specifically. The stored confusion matrices agree: both models concentrate their residual errors in the same place, most visibly between the Business and Sci/Tech classes of AG News.

Do the two encoders see language the same way? The accuracy gap says how often the models agree on a label, not whether they agree on the geometry underneath. We embed 496 words spanning

Table 4: Pending. The runs behind this table have not finished; src/figures.py writes it as soon as they do.

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Table 5: Pending. The runs behind this table have not finished; src/figures.py writes it as soon as they do.

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20 topics with each pretrained encoder and compare the two 768 dimensional spaces with linear centred kernel alignment [Kornblith et al., 2019], matching student block d to teacher block $2d$, the layer it was initialised from. Alignment is near perfect at the embeddings (0.977), still high at mid depth (0.838 for student block 3 against teacher block 6) and falls to 0.578 at the top, so distillation preserved the lower and middle geometry and let the task specialised upper layers drift. Reading words from the layer averaged, mean centred space, 7.3 of the ten nearest neighbours BERT assigns to a word are also chosen by DistilBERT, and 72 percent of DistilBERT neighbours share the topic of their query word against 69 percent for BERT, so the smaller model is marginally tidier here. An interactive version ships with the code.

5 Conclusion

DistilBERT keeps 99.2 percent of the accuracy of BERT base on four datasets while holding 39 percent fewer parameters, reaching 2.00 times the inference throughput and training in 44 percent less GPU memory, and two epochs is enough for both. The classifier is not a useful place to spend that budget: head width and depth move F1 macro by at most [pending] points across seven variants, while freezing the transformer costs [pending]. Alignment between the two encoders is near perfect at the embeddings and decays with depth, so what distillation gave up is capacity in the upper layers, not the organisation of the space. Under a latency or memory budget DistilBERT is the right default here, and the budget is better spent unfreezing the encoder than enlarging the classifier. One caveat: Yelp was subsampled and each run used one seed, so differences under half a point are noise.

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